Bilkent University Electrical and Electronics Department

EE102-01 Lab 1 Report:

Introduction to Digital Oscilloscopes

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Purpose of the experiment

The aim of this experiment is to adopt basic skills on using lab equipment. Basic functionalities of breadboards, signal generators, and oscilloscopes are learnt. Moreover, in the sixth part of experiment, the feature of capacitors is observed that there is a phase difference due to capacitive reactance.

Methodology

In this experiment, the oscilloscope, synthesized function generator and probe are mostly used equipment. In the first task, we compensated our probe with its adjustment tool. In the second task, we used the oscilloscope to monitor a 5 Vpp 1 kHz sinusoidal signal with positive and negative edge triggering. In the third task, we examined effect of turning the trigger knob on the oscilloscope with a 1 Vpp 2 kHz triangular wave. In the fourth task, we tried the acquisition modes with a 1 Vpp voltage and 5 kHz frequency square wave on the oscilloscope. In the fifth task, a 2 Vpp sinusoidal signal with frequency 1 kHz and DC offset of 1 V is applied and observed through DC and AC coupling. In the final task, an RC circuit implemented on breadboard is used to demonstrate and measure the phase difference due to the capacitor of 1 µF by applying a 2 Vpp sinusoidal signal with 1 kHz and 100 kHz frequency.

Results

Task 1:

In order to get accurate measurements, the probe has to be compensated which is matching electrical features of the probe to a specific oscilloscope. With its adjustment tool, it is screwed or loosened to make compensation. Under-compensation, over-compensation and a compensated signal can be seen at Figure 1.1, Figure 1.2 and Figure 1.3 respectively.

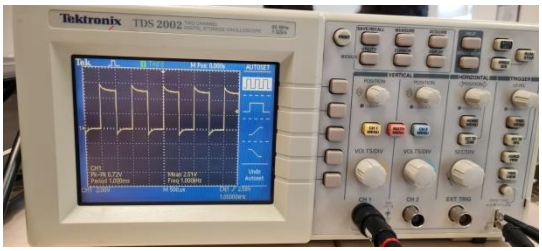

Figure 1.1

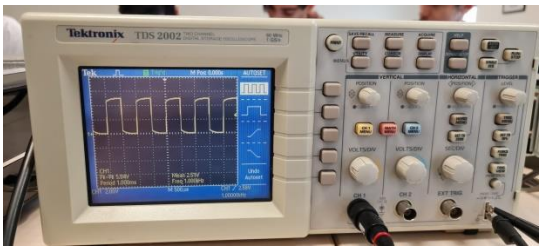

Figure 1.2

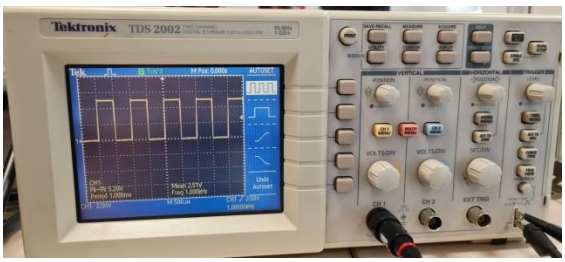

Figure 1.3

Task 2:

The slope control decides the wave part corresponding to the trigger point to be the rising or the falling edge of the signal. With 5 Vpp 1 kHz sinusoidal signal, rising (positive) edge triggering (Figure 2.1) and falling (negative) edge triggering (Figure 2.2) is applied one by one.

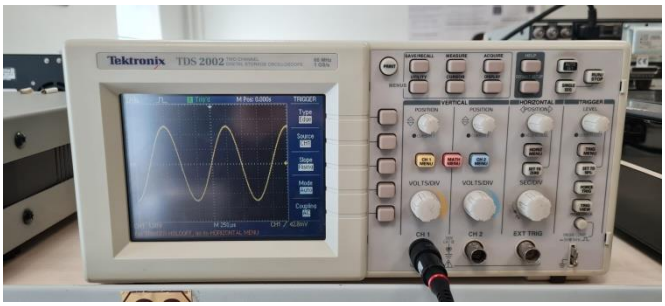

Figure 2.1

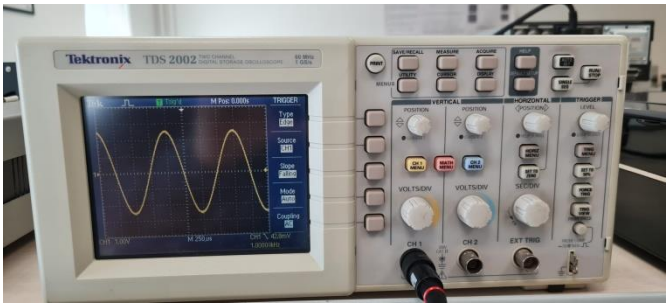

Figure 2.2

Task 3:

The trigger control enables repeating waveforms to display a captured stabile signal. The knob of triggering level changes where the trigger point is. A 1 Vpp triangular wave with 2 kHz frequency is applied through a function generator and observed on the oscilloscope. It is observed that the signals shift when trigger knob turned. If it is on the range of wave, it adjusts the appearance of the signal so that the intersection of the waveform and trigger level is horizontally centered on the screen (Figure 3.1, Figure 3.2, Figure 3.3). Otherwise, the oscilloscope cannot get the expected value so it starts moving to find a value (Figure 3.4).

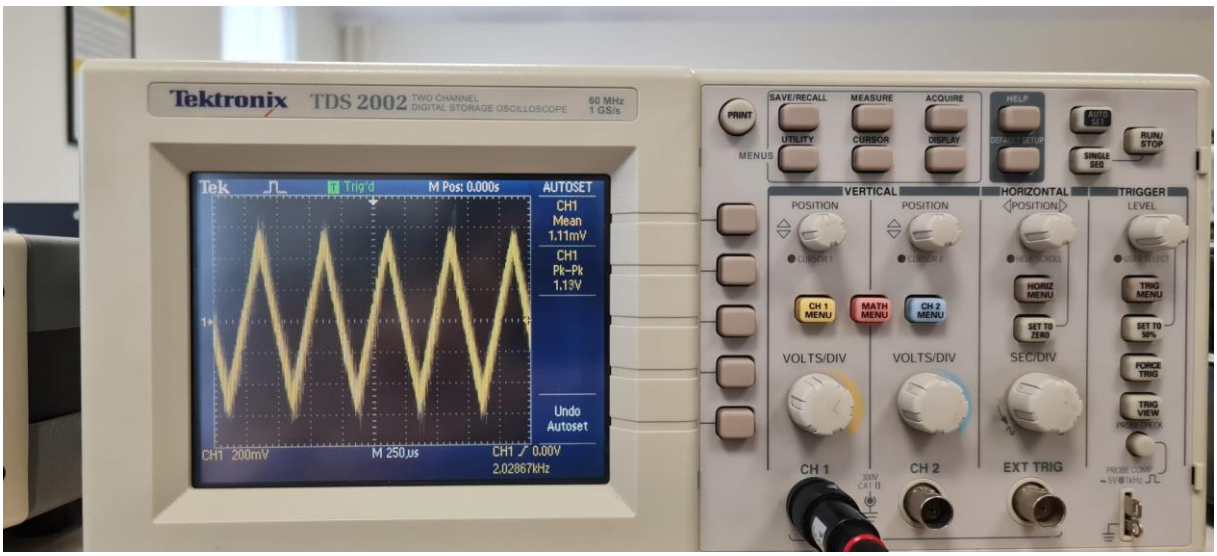

Figure 3.1

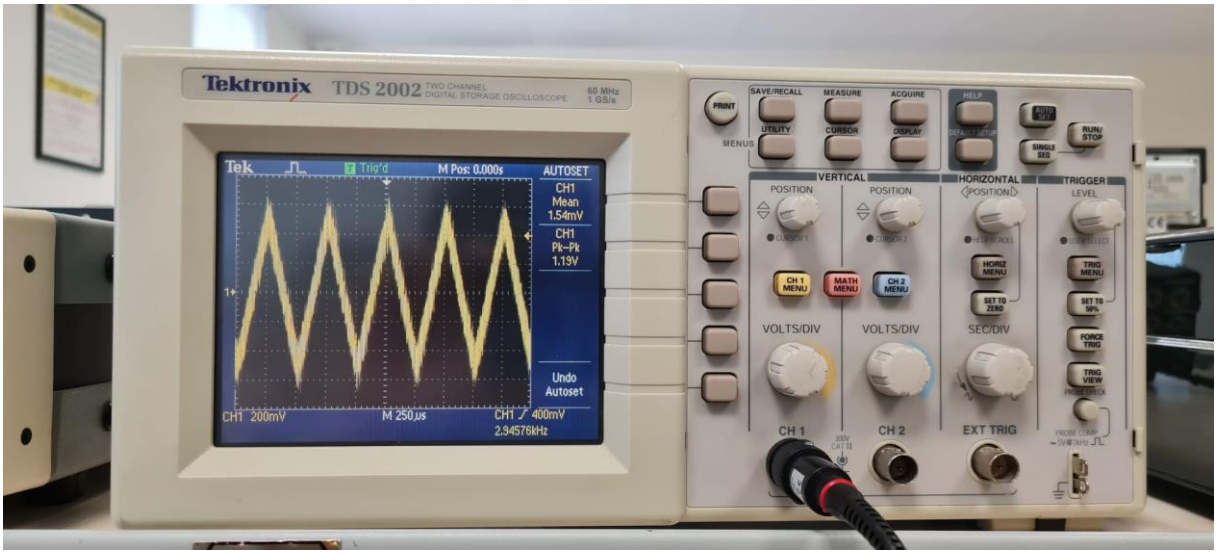

Figure 3.2

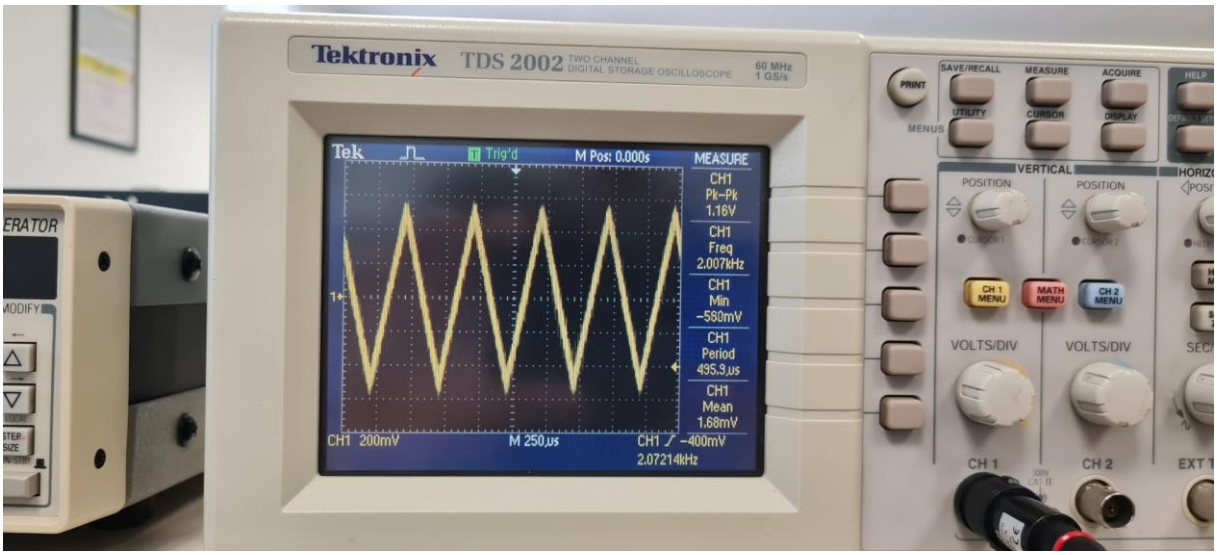

Figure 3.3

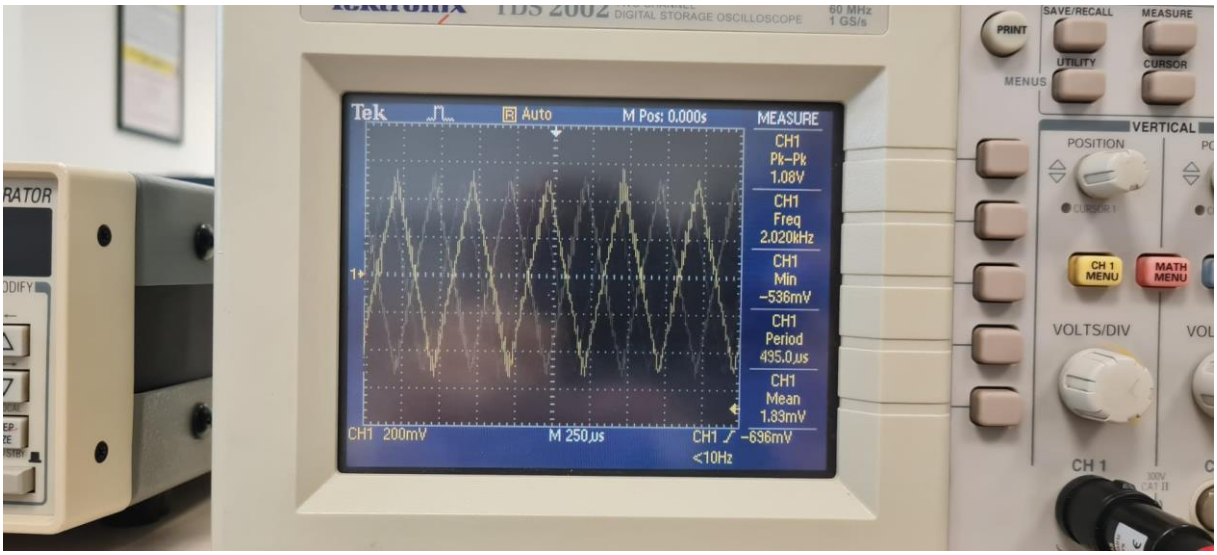

Figure 3.4

Task 4:

Digital to Analog Converter (DAC) turns digital inputs to analog signals. Analog to Digital Converters (ADC) is used for other way around. For example, the digital oscilloscope uses digital information to give image on screen which is converted through ADC from analog signal samples. Generated a 1 Vpp voltage and 5 kHz frequency square wave, the different acquisition modes are tested on the oscilloscope. In the sample mode, the oscilloscope constructs the waveform by merging recorded data from various intervals (Figure 4.1).The peak detect mode samples different max and min points and combines them (Figure 4.2). The average mode takes average of given number of inputs and draws from them so increasing the sample number causes a better image (Figure 4.3, Figure 4.4).

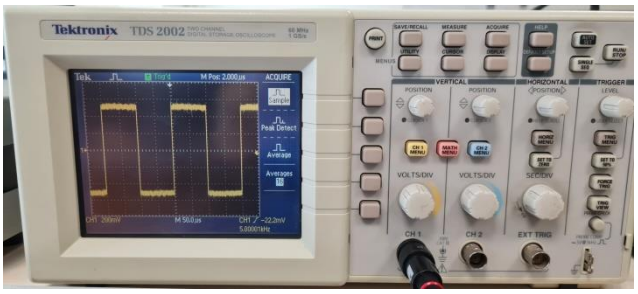

Figure 4.1

Figure 4.2

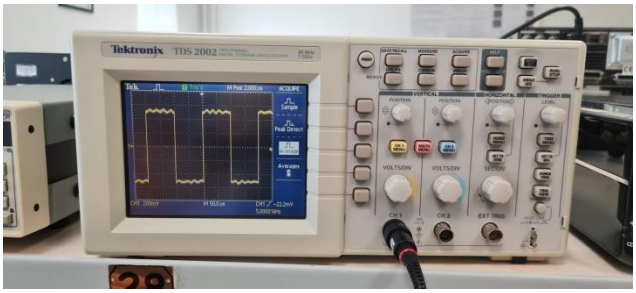

Figure 4.3

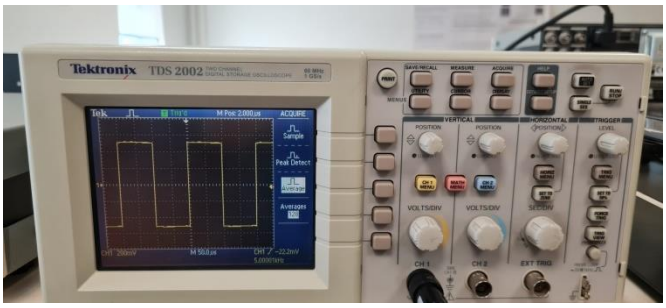

Figure 4.4

Task 5:

Through a signal generator, a 2 Vpp sinusoidal signal with frequency 1 kHz and DC offset of 1 V is applied. DC offset shifts up whole waveform by a DC value. The DC coupling mode measure mere voltage values (Figure 5.1), but the AC coupling mode measures the alterations between voltage values (Figure 5.2). Therefore, the AC coupling mode it ignores the DC offset which allow us to see change.

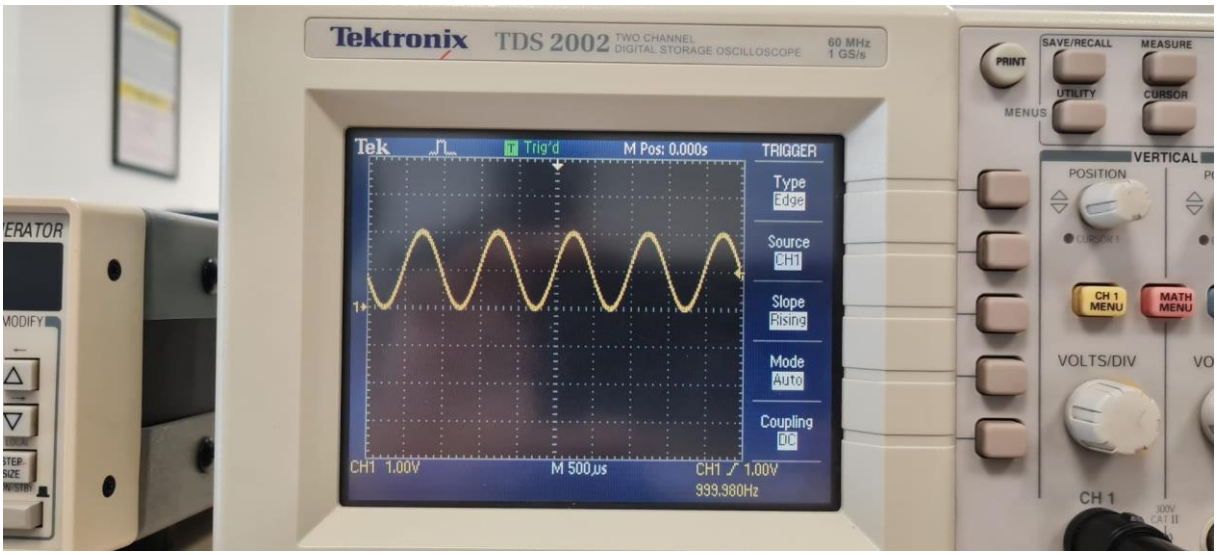

Figure 5.1

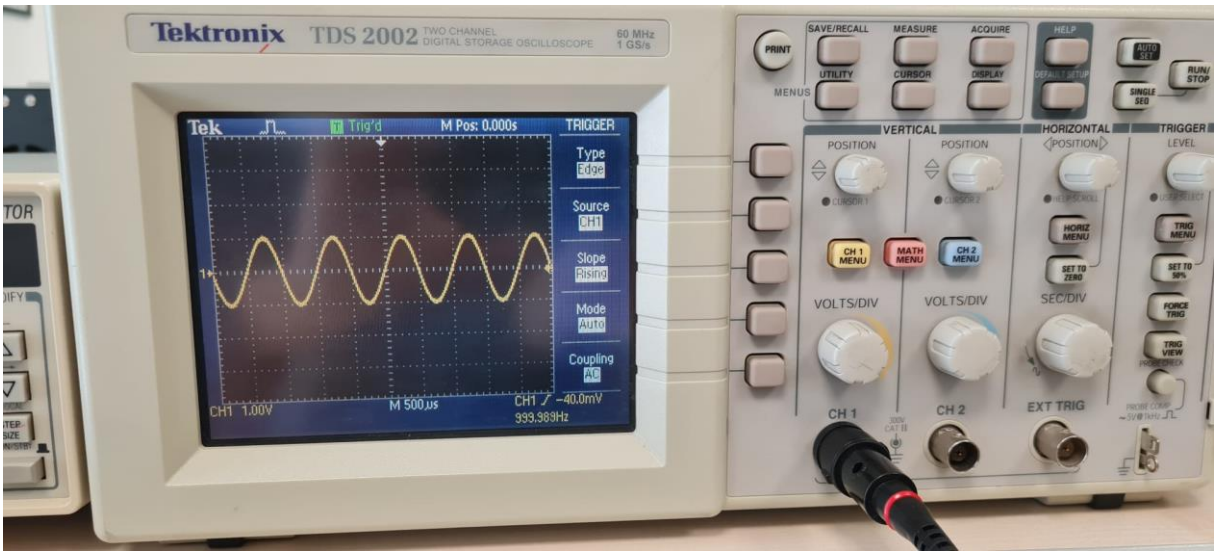

Figure 5.2

Task 6:

Breadboards are tools to utilize better demonstration and easier implementations of circuits. They have several separated metal conductive plates under the plastics. In this final task, an RC circuit is implemented on breadboard (Figure 6.1). The ground of signal generator, probes and one tip of resistor are merged to be common ground. A 2 Vpp sinusoidal signal with 1 kHz and 100 kHz frequency applied and sent to the oscilloscope through probe X. It is also connected to the capacitor of 1 µF. After that it goes to both the resistor and the oscilloscope through probe Y. Time difference between two signals coming from X and Y probes is measured using the cursor feature of the oscilloscope. Time differences are as shown in for 1 kHz and 100 kHz signal respectively. We got 28microseconds and 10 nanoseconds for 1 kHz and 100 kHz frequency, respectively (Figure 6.2, Figure 6.3). Since the formula for finding the phase difference by using Δt time difference is φ=360°.f.Δ , we got 9.97° and 0.39° for 1 kHz and 100 kHz frequency, respectively.

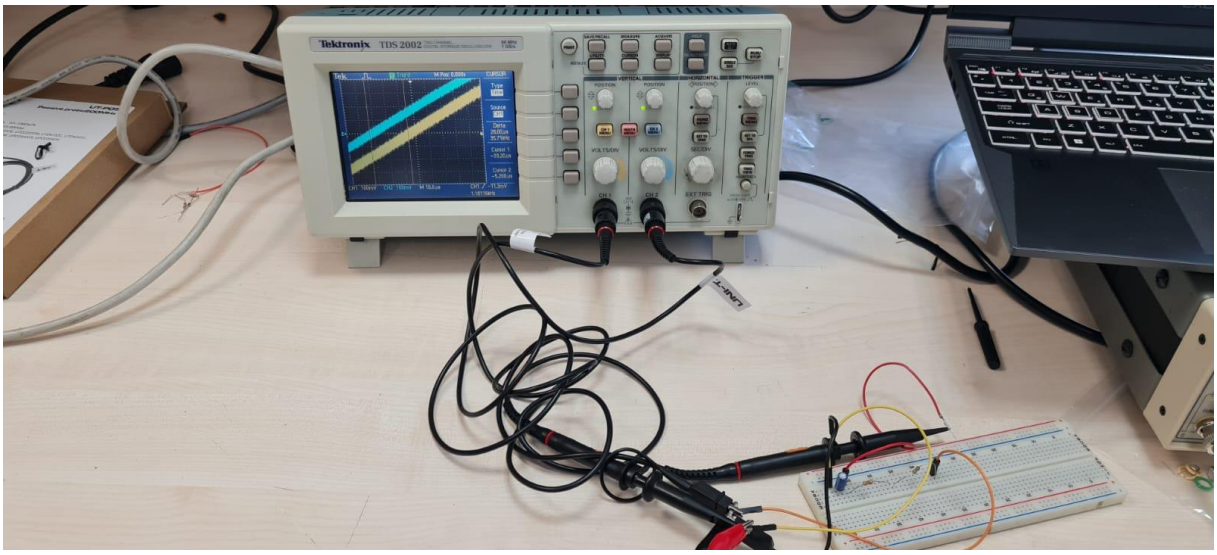

Figure 6.1

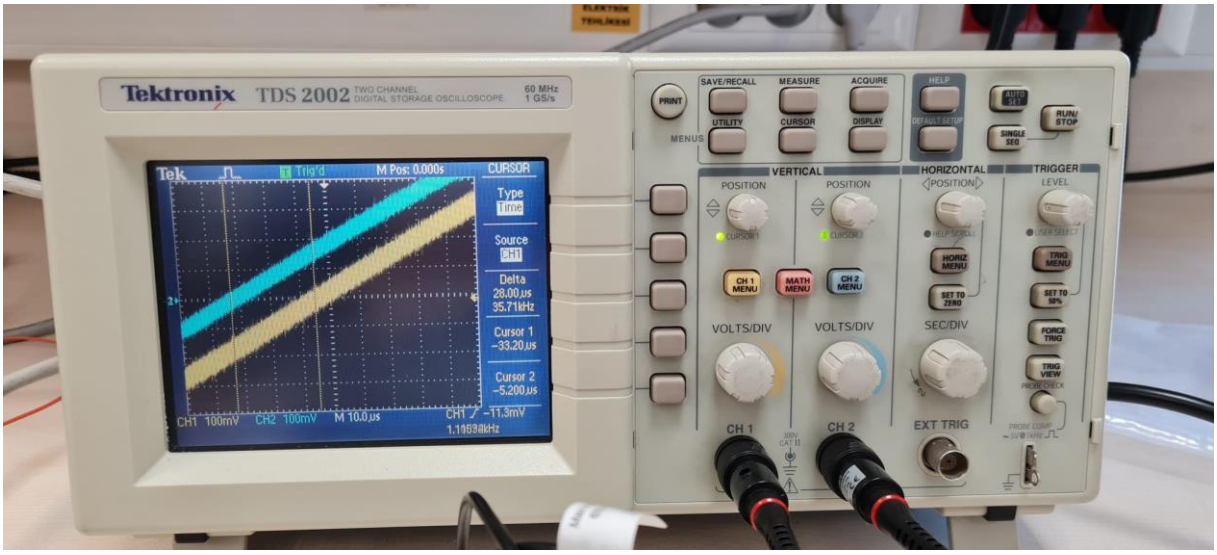

Figure 6.2

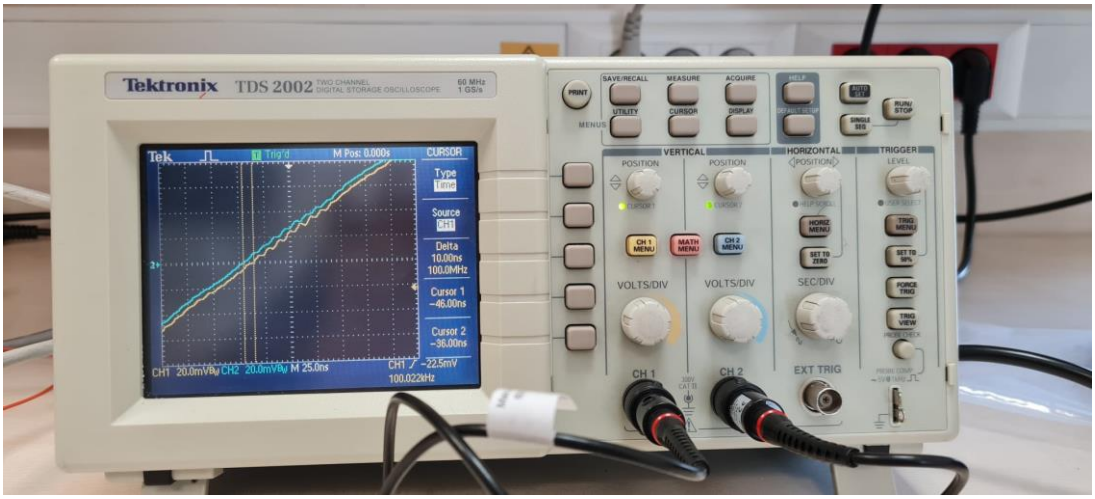

Figure 6.3

Conclusion

The experiment successfully demonstrated the aim which is learning basic functionalities on oscilloscope and signal generator. Furthermore, we showed that phase difference due to capacitive reactance changes with signal's frequency.